

Retail Sales Performance Analysis

Portfolio-Grade Data Analyst Project Report

Project Overview

This project focuses on analyzing retail sales data to uncover insights related to sales performance, profitability, customer behavior, product performance, and regional trends using Python, MySQL, and Power BI.

Business Problem

The objective was to identify high-performing categories, products, customer segments, and regions while understanding how discounts impact profitability and business growth.

Dataset Information

Retail Superstore dataset containing approximately 9,994 transactions with customer, product, sales, profit, discount, shipping, and regional information.

Tools & Technologies

Python (Pandas, NumPy), MySQL (Business Analysis & KPIs), Power BI (Dashboarding & Visualization).

Data Preparation

Performed data cleaning, missing value validation, duplicate checks, date formatting, and dataset preparation for business analysis.

SQL Business Analysis

Conducted revenue analysis, profit analysis, category performance analysis, customer analysis, regional analysis, product analysis, and discount impact analysis.

Executive KPI Summary

Total Sales: \$2.30M | Total Profit: \$286.40K | Total Orders: 5K+ | Profit Margin: 5.23% | Avg Discount: 15.62% | Avg Shipping Days: 3.96

Key Insight 1

Technology was the highest-performing category, generating approximately \$840K in sales and leading profitability.

Key Insight 2

Consumer customers contributed the largest share of sales and profit, making them the most valuable customer segment.

Key Insight 3

The West region delivered the strongest sales and profit performance among all regions.

Key Insight 4

Higher discounts negatively impacted profitability, indicating the need for optimized pricing strategies.

Key Insight 5

Top-performing products such as Canon imageCLASS contributed significantly to overall revenue.

Business Recommendations

Increase investment in Technology products, optimize discount strategies, strengthen customer retention, improve low-performing regions, and expand cross-selling initiatives.

Skills Demonstrated

Data Cleaning, EDA, SQL Query Writing, KPI Analysis, Data Visualization, Dashboard Development, Business Intelligence, Data Storytelling, and Analytical Thinking.

Project Outcome

Successfully transformed raw retail sales data into actionable business insights and developed an executive dashboard supporting data-driven decision-making.

Resume Project Description

Developed an end-to-end Retail Sales Performance Analysis solution using Python, MySQL, and Power BI, analyzing 9,994 transactions and \$2.3M in sales to identify profitability drivers, customer behavior trends, regional performance, and growth opportunities.